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Applicant Summary Information

Host Organisation: University College London

Partner Organisation (if applicable):

Strategic Theme: N/A / Health data science / Public health / Mental health / Social care

Partnership Fellowship: N/A

WTE%: 1.00

Duration: 36 months

Research Title:

Research Type (can select multiple): Clinical trial (phase 2) / Clinical trial (phase 3a – efficacy) / Clinical trial (phase 3b – effectiveness) / Clinical trial (phase 4) / Feasibility Study / Pilot study / Secondary data analysis / Cohort study / Epidemiological / Meta analysis / Qualitative study / Retrospective review / Survey / Evidence Synthesis / Evaluation / Other (please specify)

Proposed start date: 01/10/2025

Applicant Research Background

Applicant Professional Background: Not a Health or Care Practitioner

Profession: Basic Scientist / Behavioural Scientist / Biomedical Engineer / Computer Scientist / Epidemiologist / Food Scientist / Health Economist / Health Services Researcher / Information Specialist / NHS Manager / Other Research Scientist / Psychologist / Statistician / Other

Have you already registered for a PhD? No

Has this application, or a similar application, previously been submitted to this or any other funding body? Yes / No

If yes, provide details:

NIHR Pre-Doctoral Fellowship (Round 6) - Application was deemed fundable, however, not ranked highly enough to be funded with the monies at the Programme's disposal. The current application for a Doctoral Fellowship expands on the research project and training plan initially

proposed as part of the Pre-Doctoral Fellowship application. This application has also incorporated the feedback received from the reviewers. Specifically, this application elaborates on the impact this fellowship will have on my career, and the support provided by the supervisors and the host institution.

Please describe your research career to date (1000 words)

Previous Experience:

I am an early-career researcher with a strong interest in improving evidence synthesis in women's health. I have a BSc in Physiology (McGill University), an MSc in Population Health (UCL), and experience working on a variety of health research projects.

As an undergraduate, I spent two summers as a research assistant with Dr Rosemary Martino, University of Toronto. Initially, I evaluated abstracts for inclusion in a systematic review assessing the benefits of dysphagia screening in stroke patients. This introduced me to systematic reviews and gave me insight into their importance in healthcare research. I subsequently helped develop an online mixed-methods survey of speech-language pathologist practices in paediatric dysphagia; developing skills in SPSS by tabulating survey response frequencies, means and standard deviations (REF).

During undergraduate courses in reproductive physiology, I was struck by the paucity of evidence in areas of women's health. I sought work on a cross-sectional study investigating the relationship between hypertensive disorders of pregnancy and postpartum depression. Under Dr Natalie Dayan (Research Institute, McGill University Health Centre), I gained first-hand experience extracting clinical data from medical records into RedCap for analysis. Importantly, this project introduced me to women's health research and furthered my interests in this area. I have remained involved in this study as a manuscript co-author.

Prior to my Masters, I worked with Dr Samira Mubareka (Sunnybrook Research Institute) as a part of the Royal Society of Canada's One Health Working Group. I participated in group meetings and helped to develop and write a policy brief about a One Health approach to zoonoses (REF). This advanced my writing skills, and I gained experience collaborating with a large multi-disciplinary team, including a variety of experts, as well as representatives from government and Canada's Indigenous communities. This work taught me the importance of and how to synthesise existing literature to inform policy.

My MSc in Population Health introduced me to study designs used in medical research, including clinical trials and systematic reviews, and statistical approaches like regression analyses. I also gained valuable experience designing and conducting analyses in my dissertation where I investigated sex differences in the relationship between blood pressure and depression over the life course, using data from the 1946 British Birth Cohort. I have continued to collaborate on this project to produce a manuscript for publication and have presented preliminary results at a UCL departmental meeting. Completion of this degree, and particularly the dissertation, further motivated me to pursue a career in women's health.

Following my master's degree, I joined the London School of Hygiene & Tropical Medicine Clinical Trials Unit as Assistant Trial Manager for the international I'M WOMAN trial. Working on this trial

further solidified my interest in women's health research and showed me the huge impact research can have on the lives of women. It gave me greater insight and experience in clinical trial design and implementation, writing ethics applications, and recruiting trial sites. In addition, I learned important transferrable skills through managing multiple tasks simultaneously and effectively resolving unexpected circumstances that can arise during a study.

Current Employment:

In November 2023, I started my current position (Research Assistant, Meta-Analysis Group, MRC Clinical Trials Unit at UCL), funded by a National Institute of Health Research Health Technology Assessment Award (NIHR153187), to synthesise results from clinical trials which have assessed treatments for heavy menstrual bleeding. This role has combined my research interests in women's health and evidence synthesis.

My duties have included developing protocols, conducting literature searches, identifying eligible clinical trials, developing data extraction forms, and completing data extraction across 120 eligible trials. Importantly, I helped to host a workshop involving women with lived experience of heavy menstrual bleeding to inform the design of a mixed-methods survey to understand the impact of heavy periods on quality of life and women's experiences seeking treatment. I subsequently supervised MSc students to seek ethical approval, develop and analyse this survey, a manuscript for which is in preparation. I have also supervised students to undertake risk of bias assessments and have overseen a visiting student to assess trustworthiness of the trials. I have worked with the lead statistician to develop an analysis plan and establish meta-analyses of the primary outcomes. Results from these meta-analyses are anticipated in May 2025. Complexities that have arisen in this process have been a key motivation for developing my Fellowship proposal.

In early 2024, I was central in organising a Menstrual Health Research Day, to inform women, patient advocates and women's health charity representatives (e.g. Wellbeing of Women, Menstrual Health Project) about our research. During the event, I presented our research plans and sought input from attendees through a live survey and discussion regarding which outcomes should be prioritised.

I was also invited to co-author a WHO-commissioned Cochrane review comparing intravenous versus oral iron in treating iron-deficiency anaemia in pregnancy (REF). As first author, I identified and evaluated relevant clinical trials, collected and meta-analysed data, completed risk of bias, trustworthiness and GRADE assessments for each outcome included in the review, as well as contributed to writing the manuscript. From this, I gained extensive experience in every stage of conducting systematic reviews and meta-analyses and expanded my network among other researchers.

Impact of the Fellowship

This fellowship will be an important step towards my long-term career aim to become a leading methodologist and researcher in women's health.

Through my proposed research plan, formal trainings, and with support from my supervisors and collaborators, I will further my understanding of more complex research methods in evidence synthesis, including statistical and qualitative methodologies. The fellowship will also enable me to

expand my research networks within and outside academia, through proposed research visits, collaborations and attending conferences. It will also enable me to communicate results of my research, through publications and conference presentations, to establish a track record in this area. Importantly, I will develop the skills needed for me to become an independent researcher and progress as a leader in evidence synthesis for women's health.

Contextual Factors

N/A

Plain English Summary of Research (600 words)

Background Information

One in four women will have heavy periods, which means they lose a lot of blood during their period or have a period that lasts a long time. They might also feel tired, have pain and need to miss days of school or work. These things can all make women's lives harder.

Researchers have tested treatments for women with heavy periods in clinical trials. In these clinical trials, researchers measure how much blood a woman loses on her period before she has treatment and after she has treatment to see if there is a difference in the amount of blood she loses. Researchers can measure blood loss in a laboratory, or women can record how their used pads or tampons look using picture charts, or answer questionnaires.

Researchers bring together information from all these trials in research called a meta-analysis, which is a good way to understand which treatments work best. But because blood loss has been measured in different ways in each clinical trial, it is hard to do a meta-analysis. That means it has been hard to find out if some treatments work better than others.

Aim and Objectives

In the first part of my research, I will figure out how to best bring together the results from clinical trials to get better evidence about which treatments reduce bleeding the most. Then I will do research to understand what doctors, researchers and women think about how blood loss is measured in clinical trials. This will help researchers do better clinical trials in the future.

Research Methods

I will use a mixture of different research methods during this project. For the first part of my project, I will carry out a thorough review of all the clinical trials to find all the ways blood loss has been measured. I will then try to use different statistical methods to find better ways to combine results from clinical trials that have tested treatments for heavy periods.
For the second part I will speak to women, and groups of doctors and researchers to find out their opinions on ways of measuring blood loss in clinical trials.

Patient and Public Involvement (PPI)

I have spoken to four women who have lived with heavy periods about my research plans. They agree it is important to do this research so that women with heavy periods get the right treatment. They noted that it would be important to speak to women of different ages, races, or cultural backgrounds to understand different opinions on being in clinical trials that measure blood loss. Therefore, I have planned how to include a diverse group of women in my research interviews.

I plan to set up a PPI group of six women with experience of heavy periods. I will meet with this group twice each year to share my progress, get their feedback on what I have done, and advice for my next steps. They will help me to recruit women for my research interviews and give me advice on what questions I should ask. They will also help me to present my findings so that members of the public can understand them and find them helpful.

Sharing Results

I will present my research at conferences and write papers for medical journals. I will also communicate my results more widely, including using social media, blogs, or podcasts so that people affected by heavy periods hear them. I will ask advice on the best ways to do this from my PPI group.

Scientific Abstract (500 words)

Background

Heavy menstrual bleeding (HMB) affects one in four women of reproductive age. HMB is a life altering symptom which impacts all aspects of women's quality of life. Alongside the physical symptoms, women with HMB also often experience isolation from work and social activities due to fears of bleeding through clothing.

There are many causes of HMB, however, many women also suffer from HMB with no identified pathology. There are numerous treatments available to women with HMB, but there is limited high quality evidence comparing treatment options. Although many clinical trials have compared different treatments, there is large variability in how outcomes from these trials are reported. For example, some HMB trials measure menstrual blood loss (MBL) using the alkaline-hematin method or variations of pictorial blood assessment charts, while other trials report subjective measures of MBL. This poses challenges for evidence synthesis, as oftentimes many trials are excluded from meta-analyses.

A Core Outcome Set (COS) was published for HMB trials which identified nine outcomes to be collected in future HMB trials. One of the outcomes identified in the COS was menstrual blood loss (MBL). However, the COS did not specify which measurement tools should be used to measure MBL, or which timepoints should be reported. Therefore, there is still a significant need for further work to be done to standardise and improve meta-analyses of the MBL outcome in HMB trials.

Research Question

How can we improve meta-analyses of MBL outcomes in clinical trials comparing treatment options for HMB?

Objectives

1. To critically assess the measurement tools used to assess MBL in clinical trials.
2. To explore statistical approaches to improve meta-analysis for the different MBL outcomes reported.
3. To determine what is acceptable and prioritised to women and trialists when measuring MBL in HMB trials.

Methods and Timelines for Delivery

Work Package 1 (Months 1-8) - Conducting a systematic review of the literature to identify different measurement tools used to assess MBL in clinical trials and critically evaluating these tools.

Work Package 2 (Months 9-18) - Exploring statistical approaches for meta-analyses and applying these so that more clinical trial data is included in meta-analyses of the MBL outcome.

Work Package 3 (Months 19-30) - Conducting interviews and focus groups with women and trial teams to determine which aspects of MBL measures are acceptable and prioritised by different stakeholders.

Anticipated Impact and Dissemination

The proposed research will inform the choice of MBL measurement tools in future trials and will provide better estimates of the effect of treatments on MBL using data from past clinical trials. The findings of each work package will be disseminated in scientific journals and presented at relevant conferences. The PPI committee will be crucial in advising how to communicate findings in a way that is easily understandable to the public. These could include social media posts, blogs and podcasts to disseminate findings to the wider public.

Detailed Research Plan (5000 words)

What is the problem being addressed?

Despite many clinical trials, systematic reviews and a recently developed core outcome set (COS), evidence regarding which treatments are most effective for women (and those born female) who experience heavy menstrual bleeding (HMB) remains limited.

This is because trials to date have reported results inconsistently, or evaluated treatments using different outcome measures, which poses challenges for evidence synthesis. The COS provided a list of nine outcomes that it recommended should be implemented in future HMB trials. However, it did not specify how these key outcomes like menstrual blood loss (MBL) should be measured, meaning that complexity in the synthesis of trial results persists. Therefore, further work is vital to standardise outcome measurement and prevent inconsistencies in HMB trials going forward.

My proposed research aims to improve meta-analyses of MBL outcomes from HMB trials to better inform treatment guidelines and help improve the lives women suffering with HMB. I will explore how statistical methods might be used to better combine MBL data from existing HMB trials in meta-analyses. I will also determine the priorities of women and trialists when measuring MBL to inform consistent MBL measurement in future HMB trials.

Why is this research important in terms of improving the health and/or wellbeing of the public and/or to patients/service users, carers, and health and care services?

An estimated one in four women of reproductive age are affected by heavy menstrual bleeding (HMB)(REF). Women with HMB suffer lower quality of life due to both physical symptoms as well as restrictions on their social and work lives. Women with menstrual health conditions make up a key proportion of the 755,000 women waiting for gynaecology services in the UK, with many women waiting over a year for treatment (REF). These delays in diagnosis and treatment leave women suffering worsening symptoms and poor quality of life for extended periods of time. This can also lead to more complex treatments being needed when they are eventually seen by gynaecology services. Therefore, it is crucial that HMB research is prioritised now.

While there are many medical and surgical treatment options available to treat those who experience HMB, high-quality evidence synthesis to guide policy and practice is limited. The women we have spoken to have stressed the need for better quality evidence as currently many women try various treatments before finding one which is effective. The evidence of treatment effectiveness has been hindered in a large part by inconsistency of outcome measurements used across relevant clinical trials. For example, menstrual blood loss (MBL) and quality of life (QoL) have been evaluated using a range of instruments, and at a variety of time-points, which complicates meta-analysis and reduces reliability of findings.

Therefore, this research is vitally important to people affected by HMB. The proposed research will provide more reliable evidence about which treatments are most effective in reducing MBL and reduce research waste from trials that currently are not contributing to evidence syntheses. It will also explore whether reduced MBL leads to improvement in other outcomes that are important to women, including QoL. This is based on feedback from PPI groups that reduced MBL is only important if it also results in improved QoL. Lastly, this project will provide foundations for the development of a measure or tool that can be used to reliably measure MBL in clinical trials, taking the needs of women (including women from different age-groups, regions and ethnicities), trialists, and other stakeholders (e.g. guideline developers, regulatory authorities) into account. The current measures of MBL used in clinical trials can be burdensome, and it is unclear whether these measures are acceptable to all women. Therefore, this research might greatly improve women's participation in future trials, and the applicability of future trials and evidence syntheses. Ultimately, this will facilitate efficient and effective treatment strategies for women suffering from HMB and greatly improve their quality of life.

Review of the existing evidence – how does the existing literature support this proposal?

What is known about heavy menstrual bleeding (HMB) and how does it impact on those affected?

Approximately one in four women of reproductive age experience HMB, which accounts for approximately 20% of gynaecology referrals in the NHS (REFs). It is also likely the prevalence of HMB has been underestimated due to normalisation of symptoms and social stigma surrounding menstruation. There are many possible causes of HMB which can be classified using the FIGO classification system 2 (PALM-COEIN) (REF). The FIGO classification system categorises causes into either structural causes (**P**olyp, **A**denomyosis, **L**eiomyoma, **M**alignancy and hyperplasia) or non-structural causes (**C**oagulopathy, **O**vulatory dysfunction, **E**ndometrial, **I**atrogenic, **N**ot yet classified) (REF). However, the mechanisms behind which many of these causes lead to HMB is still not understood.

HMB is a symptom with life altering implications for women who suffer from it. For example, HMB is associated with poorer self-rated physical health with women experiencing symptoms of shortness of breath, weakness, and painful periods (Sinharoy 2023, Karlsson 2013). HMB is also a leading cause of iron deficiency and iron deficiency anaemia in women (Munro 2023). Alongside physical symptoms, women often need to plan activities around their bleeding due to the frequent need to change menstrual products and fear of embarrassment associated with bleeding through menstrual products (Karlsson 2013). The variety of limitations that women experience because of HMB have significant negative impacts on their social life and emotional wellbeing. There are also significant economic costs of HMB, with estimations that absenteeism or presenteeism at work due to HMB costs the UK about £11 billion per year (REF).

In research settings and previous clinical guidelines, HMB has been defined as menstrual blood loss (MBL) >80ml or a duration of >7days. However, more recently, the National Institute for Health and Care Excellence (NICE) has changed guidelines to define HMB as excessive menstrual blood loss which interferes with a woman's physical, social, emotional and/or material quality of life (REF). Previous studies suggest that many women only seek treatment for HMB when they fail to manage symptoms themselves due to the feeling that their symptoms will be dismissed by medical professionals as 'trivial' (Santer 2008). There are also disparities in prevalence of HMB in different populations. For example, many women begin to experience HMB in perimenopause (REF), and certain conditions which cause HMB, such as uterine fibroids, are more common in Black women (REF).

What are the limitations regarding evidence for HMB treatments?

There are many different treatment options available to women suffering from HMB. These include medical and hormonal treatments, such as contraceptives or tranexamic acid, as well as procedures or surgeries, such as endometrial ablation or hysterectomy. Since many treatments result in infertility or are contraceptive in nature, women who wish for future children or who are actively trying to conceive may have limited treatment options. Therefore, there is a need for treatments options to be assessed using a life course approach to reflect changes in women's treatment preferences throughout their lifetime.

Current treatment guidance in the UK for those with no identified pathology recommends using a levonorgestrel-releasing intrauterine system (LNG-IUS) as first-line treatment based (REF). However, the evidence underpinning treatment recommendations is of low certainty for outcomes such as patient satisfaction and quality of life. High-quality evidence comparing treatments in women of different demographic groups or with different causes of HMB is needed. This current lack of evidence for treatments in different populations often leads women to trying many treatment options before finding one which is effective. This leads to long time suffering for women with HMB, with many eventually opting for drastic treatments such as hysterectomies (REF).

One major reason for the of lack high-quality evidence is that across the existing clinical trials which have evaluated treatments for HMB, there is significant variation in which outcomes have been reported and how they have been measured (Cooper 2022). As a result of this, a Core Outcome Set (COS) was developed which identified nine outcomes which should be reported in future HMB trials (Cooper 2023). While the COS provides a standardised list of outcomes including menstrual blood loss (MBL), flooding, menstrual cycle metrics, dysmenorrhea, condition-specific and generic quality of life, adverse events, satisfaction, treatment failure and haemoglobin for trials, it does not indicate how each of these outcomes should be measured.

MBL is the most reported outcome in HMB trials, however, MBL has been assessed and reported in many ways (Cooper 2022, Herman 2016). These include quantitative measures of MBL such as the alkaline-hematin method which is a lab-based method of measuring the blood volume in women's menstrual products (REF). Many trials also use versions of pictorial blood loss assessment charts which have been developed (REFs). These pictorial charts are used by women to count the number

of menstrual products they use and how soiled these products are. However, many of these have not been validated with modern menstrual products (Magnay 2020). In addition to the quantitative measures used in trials, there are also many subjective measures of MBL such as having women rate whether their bleeding is better or worse than before treatment.

Each method has its own associated limitations, both in terms of acceptability to women, and in consistency or accuracy of the measurement. The different MBL measurement tools have been previously reviewed, (Magnay 2018), however, this review did not critically assess each tool using the strict methodology that exists for reviewing measurement tools (COSMIN), and there have been several studies published which aim to develop or validate MBL measures since the previous review from 2018.

The array of measurement tools is further compounded by trials making assessments of blood loss outcomes at a variety of timepoints and using different statistical summaries to report their findings.

Therefore, there is an urgent need for work in this area to develop guidance on how outcomes should be measured. Otherwise, the problem of inconsistent outcomes will get worse as more clinical trials are developed without clarity on how to measure outcomes.

Challenges for Evidence Synthesis

The variations in outcome measurements reported in HMB trials, particularly MBL outcomes, pose significant challenges for systematic reviews and meta-analyses. To date, meta-analyses comparing the treatments of HMB have been forced to exclude many trials because of differences in measurement tools used or timepoints reported (Bofill Rodriguez 2022). This exclusion of trials from meta-analyses results not only wasting of research but also reduces the ability to make conclusions about the effectiveness of different treatment options. This causes delays in the implementation of new effective treatments and may contribute to or exacerbate inequities in treatment for women with HMB.

Various statistical methods exist which may help incorporate more trial data into meta-analyses. These include investigations into the correlations between different measures or timepoints and incorporating meta-analytic priors based on these correlations (REF). It is also possible to incorporate trials which have reported different outcomes using multivariate meta-analysis methods (Hattle 2022, Riley 2017). However, no meta-analysis has investigated the use of these more complex meta-analysis approaches in the context of HMB trials.

What is the research question/aims and objectives?

Research Question: How to improve meta-analyses of menstrual blood loss outcomes from heavy menstrual bleeding clinical trials to facilitate evidence-based treatment?

Objectives:

1) To evaluate the different outcome measures used to compare menstrual blood loss in clinical trials.
2) To apply different statistical approaches to improve meta-analyses of menstrual blood loss measures.
3) To determine which outcome measurements of menstrual blood loss would be acceptable and prioritised by different stakeholders for future clinical trials.

Project Plan

This research project will consist of four work packages each associated with one of the aims listed above. The work packages include complementary quantitative and qualitative research components, which will be used to address the overall research question.

Work Package for Aim 1: Identifying and evaluating measurement tools used for assessing menstrual blood loss in clinical trials.

Many different measures of menstrual blood loss (MBL) have been used in clinical trials. There is a need to understand how these different measures compare and to critically appraise these measures for use in heavy menstrual bleeding (HMB) trials.

Reviewing Measures of MBL in Clinical Trials

I will systematically review the literature for all HMB randomised controlled trials which have collected MBL as an outcome. As part of my team's current work conducting a series of systematic reviews of treatments for HMB, we have identified 114 published trials which have collected MBL as an outcome (NIHR grant number), however we have not appraised the different measures of MBL used. As part of my fellowship, I will update the previous searches in CENTRAL, EMBASE, MEDLINE and trial registries to identify any new trials which have collected MBL as an outcome.

In the current series of reviews, we have only extracted MBL data from trials which reported continuous measures based on the alkaline-hematin method or pictorial charts (70 trials). For my fellowship, I will collect information on the MBL measures reported across all HMB trials identified. I will also collect information on trial dates, the recruitment rates, losses to follow up, and the characteristics of women included. I will then use this data to identify whether MBL measures used in trials have changed over time, whether they differ by population included, and whether participant recruitment or dropout rates are correlated with the MBL measure used.

Critically Appraising Available Measures of MBL

I will conduct an update to previous systematic review (Magnay 2018) to identify any new studies which have developed or validated new or existing MBL measures. I will also search for any qualitative studies which have assessed any measures of MBL. Updated searches will be conducted in CENTRAL, EMBASE, MEDLINE and clinical trial registries (Clinicaltrials.gov, WHO ICTRP).

For included studies which have developed or validated measures of MBL, I will collect data on the characteristics of the MBL measure used, the study samples, the results of the MBL measure and information on the feasibility and interpretability of each MBL measure. I will use the COSMIN Risk of Bias checklist for patient-reported outcome measures (REF) to assess the methodological quality of each study. I will also use COSMIN criteria for good measurement properties to rate whether each MBL measure is sufficient, insufficient or indeterminate (REF).

PPI Committee

I will meet with my PPI committee prior to starting data collection to seek advice on what information is important to collect from each study to properly review the measures of MBL. Towards the end of this work package, I will meet with my PPI committee to present my results and decide the best way to disseminate my findings.

Findings

The findings from this literature review will be reported to help inform recommendations for choice of MBL measures for future trials. These findings will also provide a comprehensive list of MBL measures which will be further investigated in work packages 2 and 3.

Work Package for Aim 2: Developing and applying statistical approaches to improve how MBL outcome measures are combined in meta-analyses.

Variations in how and when MBL is measured, and how it is reported, present complex challenges for evidence synthesis. Subsets of trials with consistent handling of MBL data may be analysed whilst other evidence is excluded, with clear implications for risk-of-bias and strength of evidence. Attempts to synthesise more comprehensively may result in confused or biased interpretations due to data variations not being handled appropriately. This work package will explore different statistical approaches that may help to meta-analyse data more appropriately and completely from trials where a measure of MBL is reported.

MBL Reported at Different Timepoints

Post-treatment measurement of MBL may take place at various lengths of time after randomisation, typically between 3 and 12 months. To address this, I will investigate the strength of correlation between MBL measures collected at different timepoints (for example, 3 and 6 months), and thereby assess whether it is more appropriate to analyse such data separately or together. For example, do MBL measures collected at 3 months differ significantly from those collected at 6 months, or can they be assumed to be equivalent for the purpose of meta-analysis?

MBL Measured with Different Tools

As mentioned in previous sections, there are a variety of different measurement tools which can be used to measure MBL, such as the alkaline-hematin method or pictorial blood assessment charts. Based on the different tools identified in work package 1, I will investigate within-study correlations between tools from validation studies or from HMB trials which reported using more than one

measurement of MBL. I will then explore statistical approaches, such as the use of Bayesian priors, to determine which MBL measures could be combined in meta-analyses.

Differences in MBL Effect Measures and Analysis Methods

Another issue which makes meta-analysis of the MBL outcome challenging is the variety of effect measures used in trial reports. For example, MBL may be analysed as a continuous measure, or may be dichotomised into "success" or "failure". Continuous measurement may be summarised using means, medians or percentage change from baseline. For binary (dichotomised) data, analysis may be via contingency tables or logistic regression. Data may be reported as raw differences between arms or might be adjusted for baseline differences or other prognostic factors. I will investigate existing statistical methods which have been developed to estimate continuous measures from binary measures, or which are used to convert medians to means. I will determine whether these existing statistical approaches can be validated in the context of the MBL outcome.

The Relationship Between MBL and Quality of Life (QoL)

While MBL is an important objective outcome in trials comparing treatments of HMB, we have received feedback from women with lived experience of HMB stressing that reduction in MBL is only meaningful if this relates to improved QoL. As part of this work package, I will investigate the correlations between MBL and QoL in trials which have reported both outcomes. I will then explore whether statistical methods such as multivariate meta-analysis might be used to synthesise both outcomes simultaneously to provide results which are more meaningful to women.

PPI Committee

As this work package is using more complex statistical methods, my PPI committee will be very helpful in advising whether combining this data into meta-analyses is still useful to women. I will also meet with my PPI committee to seek advice on what language is appropriate and easily understandable, and how I might visually present my results to the public.

Findings

The findings from this work package will provide insights on the more complex meta-analysis methods which can be used to assess treatment effects on MBL. This will allow more precise estimates of the effects of treatments on MBL than previous meta-analyses by including data from more trials.

Work Package for Aim 3: Determining which measures of menstrual blood loss are acceptable and prioritised by women and trial investigators.

It is necessary to understand what is most important and acceptable to all relevant stakeholders to determine which MBL measure should be used in clinical trials.

In this work package, I will conduct focus groups and interviews with women with lived experience of HMB and HMB clinical trial investigators and staff to understand which measures are acceptable to different stakeholders. I will use a mixture of online and in person discussions to better develop

my skills in conducting interviews and focus groups. The topic guides for the interviews and focus groups will be developed based on findings from my formal evaluations of different MBL measures in work package 1.

Interviews: Women with lived experience of HMB

Due to the potentially personal nature of discussing HMB and MBL, I have chosen to conduct interviews with women with lived experience of HMB instead of focus groups. I will aim to recruit 15-20 women for interviews using purposive sampling. I will examine what women find acceptable when needing to measure MBL and whether the MBL measure used in clinical trials impacts whether women choose to participate in trials. These interviews will also investigate whether the barriers to participating in HMB trials differ among women from different demographic groups.

To recruit women to interview, I plan to utilise the extensive PPI networks of Wellbeing of Women through my clinical collaborator, Prof Hilary Critchley and through my collaborators at Menstrual Health Project, Anna Cooper and Gabriella Pearson. I will also create adverts for recruitment which will be posted on the MRC CTU's social media and the NIHR People in Research website. These adverts will contain a link to a form which will collect basic data on the demographics of people interested to participate. I will then use purposive sampling to ensure that I interview women from a wide range of demographics. The demographic characteristics of the women interviewed will be included in the publication of the findings from this work package.

Focus Groups: HMB trial investigators and staff

To understand how and why trial investigators have chosen to use a specific MBL measure in their HMB trials, I will conduct focus groups with trial teams. I will aim to conduct four focus groups with trial teams each with about five participants. In the focus groups, I will examine whether trial teams considered different measures of MBL, and which factors led to their decision to use the MBL measure they chose. These discussions will also cover whether trialists experienced feasibility issues or whether they felt participant dropouts were dependent on the MBL measure collected.

To recruit trial staff, I will use the networks I have gained through working at the MRC CTU at UCL on a series of systematic reviews of treatments for HMB. While working on these reviews, I have established communications with investigators from some of the HMB trials.

Data Analysis

These focus group and interview discussions will be audio recorded and transcribed. The data from these focus groups will be analysed in NVivo using a mixture of deductive and inductive reflexive thematic analysis.

Consensus Workshop: Representatives from different stakeholder groups

If the interview and focus group findings indicate that there are differences in what MBL measures are prioritised by women versus trial staff, I will hold a consensus workshop where members from each group will discuss together. The aim of the consensus workshop would be for each group to

express their thoughts to each other and determine whether consensus about which MBL measures are most important can be reached. This will be used to help interpret the data from the interview and focus group analyses.

PPI Committee

For this work package, I will seek advice from my PPI committee to determine which demographic groups are particularly important to reach in interviews to inform sampling. I will also do a data clinic with my PPI committee where I will present some of the codes and initial themes I have found to get their thoughts on my interpretation of the data.

Findings

The findings from this work package will be used to determine what needs to be incorporated into a tool to measure MBL. This may mean that an existing measure of MBL could be recommended for use in future HMB trials or may show there is a need for development of a new tool incorporating these findings.

Dissemination, outputs and anticipated impact

Dissemination and Planned Outputs

Peer-Reviewed Publications and Conference Presentations

The findings from each work package will be submitted as articles to peer-reviewed journals and I will also attend conferences throughout my fellowship to present my findings.

Work Package 1: I will publish the critical assessments of MBL measurement tools in a relevant women's health journal to inform their use in future HMB trials. I will also aim to present these findings at a relevant conference such as International Clinical Trials Methodology Conference at the end of year 1 of my fellowship.

Work Package 2: I will publish an article based on my findings from applying complex statistical approaches to meta-analyse the MBL outcome in a relevant methodology journal to help inform their use in other meta-analyses. I will also present these findings at a relevant evidence synthesis conference, such as the Cochrane Colloquium at the end of year 2.

Work Package 3: I plan to present the findings from my focus group and interview discussions at a relevant women's health conference in year 3. I will also submit these findings for publication in a relevant women's health journal.

Disseminating Findings to the Public

I plan to share the findings of my research project with help my collaborators, Prof Hilary Critchley at Wellbeing of Women and Anna Cooper and Gabriella Pearson at the Menstrual Health Project. Both Wellbeing of Women and the Menstrual Health Project have large social media presences and

networks of women with lived experience of HMB. I will also utilise the social media accounts from the MRC CTU at UCL to make posts or podcasts to disseminate my findings.

My PPI committee will be crucial in developing plans to disseminate my findings to women with HMB and other members of the public. I will be guided by my PPI committee on how to best disseminate my findings so they are easily understandable and accessible to a wide audience.

Anticipated Impact of Research Findings

Informing Future Research

The findings of my research will have significant impact on trialists conducting future trials in women with HMB. I will present the findings of my work to the COS working group with the help of my secondary supervisor, Dr Natalie Cooper. Based on my findings, I will participate in discussions of whether an appropriate measure of MBL already exists which can be recommended for future trials. If there is no appropriate measure based on my findings, we will discuss the best ways to develop an MBL measure as part of a future body of work.

My findings from applying different statistical approaches to meta-analyse MBL outcome measures could also inform researchers of how to deal with variations in outcome reporting in future meta-analyses of HMB trials, or in other clinical areas.

Informing Guideline Developers and Regulatory Processes to Improve Health

As part of my training plan, I will do a work placement with the MHRA which will be organised with the help of my collaborator Dr Jahnavi Daru. During this work placement, I will have the opportunity to present my findings and explore how my findings might facilitate more efficient decision making within the agency around licensing of treatments.

I will also use my team's current collaborations with NICE to explore how my research findings may be incorporated into future NICE guidelines for treatment of HMB.

The Impact on Women and the NHS

Apart from the significant impacts of my findings on future research in HMB, my work to better estimate the effects of treatments on MBL and QoL will inform guidelines and medical practice. By making use of all the available trials and reducing research waste, my findings will help to efficiently provide women with effective treatment options for HMB. This will lead to significantly improved quality of life for women with HMB and will reduce the burden of HMB within the NHS.

Project management

My research project will be supervised by Dr Claire Vale (primary supervisor), Dr Annabelle South (secondary supervisor), Mr David Fisher (secondary supervisor) and Dr Natalie Cooper (secondary

supervisor). At the beginning of my fellowship, I will meet with my primary supervisor weekly and then move to meeting twice per month.

The management and duration of each work package is outlined below. A detailed Gantt chart with timelines for my research project and training plan is available in the Appendix.

Work Package 1 - Identifying and evaluating measurement tools used for assessing menstrual blood loss in clinical trials. This work package will be done in months 1-8 (duration 8 months). During this time, I will also have regular monthly meetings with my secondary supervisor Dr Natalie Cooper.

Work Package 2 - Developing and applying statistical approaches to improve how MBL outcome measures are combined in meta-analyses. This work will be conducted during months 9-18 (duration 10 months). While completing this work package, I will have monthly meetings with my secondary supervisor Mr David Fisher, who will provide guidance on applying statistical methods.

Work Package 3 - Determining which measures of menstrual blood loss are acceptable and prioritised by women, trial investigators and regulators. This work package will be done in months 19-30 (duration 12 months), however, the preparatory work including ethics approval and recruitment of participants will be done prior to month 19. During this work package, I will meet monthly with my secondary supervisor Dr Annabelle South, who will provide support on conducting qualitative research.

Writing of Thesis: I will spend the final 6 months of my fellowship (months 31-36) with a focus on the writing up my PhD thesis for submission. During this time, I will also work with my supervisors and collaborators to consider how findings from my work packages could be used to develop guidance for future research and guideline development.

Ethics

Ethics approval will be needed for work package 3 which will involve interviews with women and focus groups with trial teams. Approval will be obtained from the UCL Ethics Committee prior to beginning to recruit participants. As recruitment will not be done via the NHS, a SoECAT is not required.

Success criteria

The success of my research project will be measured by achieving the milestones associated with each work package listed below.

Work Package 1

- Successfully host first PPI Committee meeting
- Present findings of work package 1 at a relevant methodology conference such as ICTMC
- Submit findings of work package 1 for publication at a relevant peer-reviewed journal
- Develop outputs of work package 1 based on guidance from PPI Committee

Work Package 2

- Successfully host two PPI Committee meetings
- Present findings of work package 2 at a relevant evidence synthesis conference such as Cochrane Colloquium
- Submit findings of work package 2 for publication at a relevant peer-reviewed journal
- Work with PPI Committee to develop outputs so findings are understandable to the public

Work Package 3

- Successfully host two PPI Committee meetings
- Recruit and interview 15-20 women from a wide range of demographics
- Recruit and conduct focus groups with 3-4 HMB trial teams
- Present findings of work package 3 at a relevant women's health conference
- Submit findings of work package 3 for publication at a relevant peer-reviewed journal
- Develop outputs with PPI Committee to disseminate results to the public

Patient and Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal (350 words)

In developing this proposal, I have spoken with four women with lived experience of heavy menstrual bleeding to present my project ideas and seek feedback from them. Two women I have spoken to are currently steering committee members on the ongoing systematic reviews which have motivated my research proposal. As such they are well-versed with some of the complexities that have been encountered in trying to synthesise the results of trials. Two of the other women I've spoken to are the founders of Menstrual Health Project, a charity focused on supporting women with menstrual health conditions. Both women have lived experience of heavy periods, and importantly, a large national network of women with similar experience.

All the women were supportive of my proposal, and they fully appreciate the need for better evidence to inform treatment and current guidelines for clinical practice.

While they agreed that the statistical methods applied in Work Package 2 would help to provide better evidence, they stressed the need for PPI Committee advice throughout this work package to ensure that the results of these analyses are useful to women. They also identified the need for me to work with my PPI committee to find ways to creatively translate these statistical findings so they can be easily understood by the public.

Importantly, each of the women I spoke to identified the need to consider the applicability of different menstrual blood loss measures to women of all cultures and backgrounds.

Following these discussions, I have added in the need for purposive sampling of women in Work Package 3 to include a diverse group of women. I have also added to Work Package 1 that I will examine whether trial conducted in different countries have used different ways to measure menstrual blood loss.

Finally, we sought the input from these four women, as well as lay members from the RSS service to review the content, feasibility and language of the PPI section and Plain English Language Summary of this proposal. Based on their feedback, I have simplified some of the language and added explanations for terms such as 'clinical trials', 'focus groups', and 'dissemination'.

Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided (350 words)

In progress.

Training and Development and Research Support

Proposed training and development programme (1000 words)

Alongside conducting my research project, I will take undertake a variety of training activities during my fellowship. This will be a mixture of training courses to learn new skills needed for my research project, trainings to develop as a leader, attending conferences and conducting research visits or work placements. A detailed timeline of my training plan is provided in the Appendix as a Gantt Chart.

Trainings in Research Methodologies

To gain the skills needed to conduct my critical assessments of all available measures of MBL (Work Package 1), I will attend the following courses:

- 'Search strategies for literature searching' (UCL) (2 hours)
- 'Introduction to search strategies for systematic reviews in health' (UCL) (2 hours)
- 'Reading a paper: critical appraisal of a randomised controlled trial' (UCL) (2 hours)
- 'Clinimetrics: Assessing Measurement Properties of Health Measurement Instruments' (Epidm - online) (6 days)
- 'Successfully incorporating patient-reported outcomes (PROs) in clinical trials' (University of Birmingham – in person) (2 days)

To learn about the statistical methods used in more complex meta-analyses which I will explore in Work Package 2, I will attend the following courses:

- ‘Statistics for Meta-Analysis' (UCL) (4 hours)
- ‘Introduction to Network Meta-analysis' (University of Bristol - online) (2.5 days)
- ‘Multivariate meta-analysis' (University of Sheffield - online) (2 days)
- ‘Introduction to Regression Analysis’ (UCL) (1 day)
- ‘Statistical Methods for Meta-analysis of Individual Participant Data’ (University of Birmingham – online) (3 days)
- ‘Stata: an introduction’ (UCL) (2 hours)

To gain the skills I will use for my qualitative research in Work Package 3, I will attend the following courses:

- ‘Qualitative Research Methods in Health’ (UCL) (30 hours)
- ‘Introduction to NVivo for qualitative and mixed-methods analysis’ (University of Surrey – online) (2 days)
- ‘Introduction to qualitative analysis’ (UCL) (2 hours)
- ‘Introduction to qualitative analysis: Interviewing’ (UCL) (2 hours)
- ‘Focus groups in qualitative research – theory and practice’ (UCL) (2 hours)

General Skills and Leadership Trainings

Alongside the courses to gain more research methodology skills, I will also undertake a variety of courses to develop general skills which will help me to becoming an independent researcher. As a PhD student at UCL, I will have access to the UCL Doctoral Skills Development Programme (DSDP) where I will attend workshops such as:

- ‘Patient and Public Involvement’ (UCL)
- ‘Facilitation: How to get the most out of your PPI activities’ (UCL)
- ‘Writing and publishing a research paper’ (UCL)
- ‘Effective Presentations: Potential Energy’ (UCL)
- ‘Leadership development as a researcher’ (UCL)
- ‘Women in Research’ (UCL)
- ‘Academic career planning for Doctoral researchers’ (UCL)
- ‘Networking skills’ (UCL)
- ‘Building your research pipeline and visibility’ (UCL)

As a member of the Meta-Analysis Group at the MRC CTU, I will remain involved in the meta-analyses being done by the team for more informal training. This will help me to gain more experience with different stages of meta-analyses across various clinical areas and will help me to grow my network with the many collaborators of the team. I will also have opportunities to attend the weekly seminars held in the MRC CTU as well as any relevant seminars hosted by other UCL departments.

Attendance at Conferences

Throughout my fellowship, I intend to disseminate the results from each work package at a relevant conference. At the end of year 1, I will present my findings from work package 1 at a clinical trials methodology conference such as ICTMC. The results of work package 2 will be disseminated at an evidence synthesis conference like the Cochrane Colloquium. In year 3, I intend to present the findings from work package 3 at a relevant women's health conference.

Through attending these conferences, I will develop my skills in presenting my research to others. I will also have the opportunity to attend various educational workshops which will be held at each conference. Apart from the skills I will gain from attending these conferences, I will be able to grow my networks among other researchers and clinicians working in areas of clinical trials, meta-analyses and women's health.

Work Placements and Research Visits

During work package 2 of my project, I will visit Dr Jack Wilkinson at University of Manchester for one week. Dr Wilkinson has plans to apply more complex statistical approaches like multivariate meta-analysis to outcomes in stroke trials. During my visit, I will compare my findings from using different meta-analysis approaches to HMB trial with his findings from analysis of stroke trials. This sharing of findings will further develop my understanding of these complex statistical methods and will allow us to compare applicability of these methods across different clinical areas.

In work package 3 of my research project, I will conduct a one-week research visit with Prof Hilary Critchley at University of Edinburgh. Prof Critchley has conducted trials in HMB and has agreed for me to conduct some of my focus groups with her teams as part of my work package 2. While visiting University of Edinburgh, I will also attend tutorials as an observer at the medical school to learn more about menstrual disorders and their treatments. Prof Critchley has also offered to have 1-1 meetings with me where needed to mentor me and help me to develop as an independent researcher working in the field of women's health.

During year 3 of my fellowship, I plan to do a one-month work placement within MHRA with the help of my collaborator Dr Jahnavi Daru. During this work placement, I will rotate through the different departments of the MHRA to understand what the processes are for licensing and monitoring safety of medical treatments and devices. This will give me a good understanding of what regulatory agencies require from clinical trials to license treatments. While conducting my work placement, I will also disseminate my research findings to teams in the MHRA so that my findings might be incorporated into their assessments of medical treatments and devices.

Supervisors – Justification of Participation (250 words each)

Dr Claire Vale, Principal Research Fellow, UCL

Claire has supported Lily to develop this Fellowship application and is committed to providing continuing support throughout. She will review Lily's progress against key

milestones, helping her to manage timelines, and to upgrade and complete on time. Importantly, Claire will support Lily to make decisions regarding the direction of her research and manage potentially conflicting priorities. She will lead the supervisory panel.

Claire and Lily will meet weekly initially, moving to fortnightly as Lily progresses through her PhD. Claire will help Lily to set up a Thesis Committee, whose role is to provide degree-spanning, independent support and advice about academic and training progress.

Claire has supervised 2 PhD and 2 MSc students. She was appointed as Departmental Graduate Tutor (Research) for the UCL Institute of Clinical Trials and Methodology in 2018, overseeing the Institute's post-graduate research (PGR) programme and contributing to Faculty PGR oversight. She is an active member of the MRC-NIHR Trials and Methodology Research Partnership Doctoral Training Partnership (TMRP DTP) Leadership Team, supporting doctoral students across the breadth of clinical trials methodology.

Claire has led systematic reviews and meta-analyses to assess treatment effects in patients with a variety of cancers, and recently, with COVID-19. Since 2023, she has been Co-PI for NIHR-funded meta-analyses of treatments for heavy menstrual bleeding. Claire has conducted methodological research to improve the design and conduct of systematic reviews, and teaches these methods at Undergraduate and Postgraduate level. She has led work to actively involve patients and the public in meta-analysis and clinical trials.

Dr Annabelle South

Annabelle is a mixed methods researcher with experience of conducting qualitative research with both patients and staff involved in running trials. She has worked in health research for almost 2 decades. Her research on sharing trial results with participants has influenced HRA guidance on this topic. Her main research interests are participants' experiences of trials; how research is communicated to different stakeholders; and the impact of research. These fit closely with the focus of WP3, with its emphasis on the perspectives of potential trial participants. Annabelle has substantial experience of PPI in research, particularly around the communication of research findings. She currently Chairs the Unit's PPI Group. As a health research communication expert, Annabelle will also be able to advise Lily on issues around the communication of the findings of the complex statistical methods explored in WP2.

Annabelle is an experienced mentor and is currently supervising one PhD student and one pre-doctoral clinical fellow and has supervised two MSc research projects which have received distinctions. Annabelle established the Qualitative Research Group at the MRC Clinical Trials Unit at UCL, which foster qualitative research skills and collaboration among early career researchers. Annabelle is experienced at working in multi-disciplinary teams, explaining qualitative research concepts to researchers from other disciplines, which will be important when co-supervising this mixed-methods PhD.

We anticipate that Lily will meet with Annabelle monthly while working on WP3, either in person or through Microsoft Teams, with interaction between meetings taking place via email and Teams.

Mr David Fisher

Dr Natalie Cooper

Collaborations (600 words)

As a Doctoral Fellow in the MRC CTU at UCL, I will have the opportunity to collaborate with other graduate students and staff who working in clinical trials and meta-analyses. Alongside those in the UCL community, I will also work with external collaborators to expand my network and learn the skills I need to progress as an independent researcher.

Collaborator 1: Dr Jack Wilkinson, University of Manchester

Jack is a Senior Lecturer based at the Centre for Biostatistics at University of Manchester. He has extensive experience working in systematic reviews and meta-analyses with a particular interest in reproductive medicine. He is also a lead researcher in conducting assessments of trial integrity, currently leading the NIHR-funded INSPECT-SR project to develop a tool to identify problematic trials. Towards the end of Work Package 2, I will visit Jack to share my findings of applying different statistical methods to meta-analyses of MBL outcomes. Jack has similar work planned using approaches like multivariate meta-analysis for outcomes in stroke trials. This research visit will allow us to compare methodological findings applied across different clinical areas. This visit will also help to grow my network among other research methodologists working in evidence synthesis.

Collaborator 2: Prof Hilary Critchley, University of Edinburgh

Hilary is a Professor of Reproductive Medicine at the University of Edinburgh. She is a distinguished researcher with a focus on understanding the mechanisms involved in menstruation and abnormal uterine bleeding. She is also currently Chair of the Research Advisory Committee at Wellbeing of Women and ex-Chair of the Menstrual Disorders Committee of the International Federation of Gynecology and Obstetrics (FIGO). I will conduct a research visit with Hilary during Work Package 3 of my research project. During this visit, I will conduct focus groups with members of her teams who have worked on heavy menstrual bleeding clinical trials. Hilary will also act as a clinical advisor for the overall project and will attend meetings with my supervisors and I as needed. In addition, will help to build my network internationally among others working in women's health research.

Collaborator 3: Dr Jahnavi Daru, Medicines and Healthcare products Regulatory Agency (MHRA)

Jahnavi is a Women's Health Expert Assessor at the MHRA and a Consultant at the World Health Organization (WHO). In her role at the MHRA, she works in the post marketing area of product development to improve patient access to effective treatments. With the help of Jahnavi, I will do a one-month work placement at the MHRA. During this work placement, I will rotate around different departments to understand the licensing processes in the UK. This will give me great insight into how evidence syntheses can influence regulatory decisions. I will also get to present the findings of my research project to different teams in the MHRA to determine how my findings might be applied in their work.

Collaborator 4: Anna Cooper and Gabrielle Pearson, Menstrual Health Project

Anna and Gabrielle are the Co-Founders of Menstrual Health Project, a UK charity focused on supporting people with lived experience of menstrual conditions. They have both suffered from endometriosis and heavy menstrual bleeding and have had various medical and surgical treatments for these conditions. Through the charity, they run a variety of activities aimed to improve menstrual health education and to help empower patients suffering from menstrual conditions. Anna and Gabrielle will be members of my PPI committee and will collaborate on the project. Through their collaboration, I will use their wide networks and social media to recruit women to interview in work package 3 and to help disseminate my findings to the public.

Host Organisation Support Statement (1000 words)

As Director of the MRC Clinical Trials Unit (MRC CTU), and Institute of Clinical Trials and Methodology (ICTM) at UCL, I would like to support the Doctoral Fellowship application from Lily Nicholson. Lily is an early career researcher, embedded in the Meta-Analysis Programme at the MRC CTU. Her background in biomedicine and public health; prior experience as a trial manager for the I'M WOMAN trial in post-partum haemorrhage, and current research assistant position, where she is contributing to a systematic review and series of complex meta-analyses of treatments for heavy menstrual bleeding, make her an ideal candidate to pursue this doctoral training. Indeed, Lily's research proposal has been motivated and shaped by her experience working in her current post; she has first-hand experience of the complexities and challenges of meta-analysis of menstrual blood loss, and direct engagement with women with lived experience of heavy menstrual bleeding, which have fuelled her desire to improve the research landscape and treatment outcomes for women affected by heavy menstrual bleeding.

The proposed supervisory team is well-equipped to ensure the best support to Lily. They span a range of relevant quantitative and qualitative research skills and techniques in clinical trials and in meta-analysis, thus will be able to provide the hands-on training bespoke to Lily's professional development. Her primary supervisor has direct experience of seeing doctoral researchers to completion of their PhDs, as well as being the Graduate

Tutor for ICTM, where she works to provide a high-quality student experience for the Institute. Inclusion of Dr Cooper as a specialist clinical advisor, providing a direct link between Lily and prior core outcome set development work ensures Lily's work will have applied clinical, as well as methodological impact. This support from supervisors, coupled with the training programme that Lily has set out, incorporating formal training and research visits with key collaborators, will ensure that she graduates from her PhD equipped with the skills and knowledge she needs to establish herself as an expert methodologist in women's health research, and in evidence synthesis more broadly. This is particularly vital at a time when women's health has been identified as a national priority for research and improvement.

Lily's Doctoral training will be hosted within the UCL Institute of Clinical Trials and Methodology (ICTM) which boasts the largest concentration of academics working in clinical trials methodology in Europe. As well as tackling internationally important health questions, and carrying out innovative and challenging studies, ICTM is a leader in training and mentoring early career researchers including PhD students, who are embedded within the work of the Institute. ICTM early career researchers, including our doctoral students, are supported by a bespoke Early Careers Network that hosts regular events and activities to support career and skills development and alerts them to relevant external training and career development events. Being embedded in ICTM also exposes early career researchers to wide-reaching networks and networking opportunities, including researchers and research teams across UCL; national networks of Clinical Trials Units, and the Trials Methodology Research Partnership and internationally, for example through vast networks of clinical and methodological collaborators, the Cochrane Collaboration and Society for Research Synthesis Methods. Our cohort of post-graduate research students, have the advantage of working and being mentored in a stimulating, supportive and research-led environment of clinicians, statisticians, scientists and operations staff. The Institute offers weekly seminars delivered by both internal and external speakers, which the Fellow will be encouraged to attend. Additional informal training opportunities including for example Methodology Work in Progress meetings, where methodologists share ongoing projects and provide peer support to resolve difficulties are also available to the Fellow to attend or to lead as an active contributor.

Specifically, Lily's fellowship will be supported within the Meta-Analysis programme at the MRC Clinical Trials Unit – the largest department within ICTM, which has world renown as a centre of excellence for the design and conduct of clinical trials, and meta-analyses, and aims to innovate the methodology that underpins these important clinical studies to improve health for all. Our meta-analysis programme has a strong reputation for delivering high impact meta-analyses, based on best practice methodology spanning 3 decades. Academics within this programme are actively involved in meta-analyses across a variety of disease areas (e.g. cancer, COVID-19, women's health and tuberculosis). They

have produced rigorous evidence that has influenced national and international guidelines and treatment of patients worldwide. They lead methodological advancements in systematic reviews, network meta-analysis and estimating treatment interactions in a meta-analysis, and developing associated software (e.g. metan in Stata). They deliver teaching and training on the conduct of systematic reviews both within UCL and externally.

UCL will further support Lily's professional development through its Doctoral Skills Development Programme. Mapped to the Researcher Development Framework, this is an extensive programme of training opportunities around engagement, influence and impact, knowledge and intellectual abilities, research governance and organisation, and personal effectiveness that all UCL doctoral students are required to undertake, in addition to mandatory training for example, in research integrity. Our Faculty (Population Health Sciences, FPHS) also requires doctoral students to establish an independent thesis committee, whose primary role is to ensure that the training and development plan is suitable and meeting the needs of each student. UCL and FPHS also provides support through induction events; annual PhD research symposia, poster, blog and 3-minute thesis competitions, Faculty student networking and social events.

UCL and the ICTM are committed to maintaining an inclusive and supportive research culture. UCL are signatories of the Concordat to Support the Career Development of Researchers, and have received an HR Excellence in Research award. UCL currently holds an institutional Silver Athena Swan award; ICTM currently holds a Bronze Athena Swan award, and will shortly be applying for a Silver award. UCL has a Bronze Race Equality Charter Award. All staff are required to undertake training in Equality, Diversity and Inclusion.